Same Mean, Different Deadline Risk

AP Statistics · Unit 2 · Topic 2.9 · TEACHER KEY

CED TETHER

- **Skill 3.B** — Determine parameters for probability distributions.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU VAR-5** — Probability distributions may be used to model variation in populations.
- **LO VAR-5.C** — Calculate parameters for a discrete random variable. [Skill 3.B]
- **EK VAR-5.C.1** — A numerical value measuring a characteristic of a population or the distribution of a random variable is known as a parameter, which is a single, fixed value.
- **EK VAR-5.C.2** — The mean, or expected value, for a discrete random variable X is $\mu_X = \sum x_i \cdot \mathrm{P}(x_i)$.
- **EK VAR-5.C.3** — The standard deviation for a discrete random variable X is $\sigma_X = \sqrt{[\sum (x_i - \mu_X)^2 \cdot \mathrm{P}(x_i)]}$.
- **LO VAR-5.D** — Interpret parameters for a discrete random variable. [Skill 4.B]
- **EK VAR-5.D.1** — Parameters for a discrete random variable should be interpreted using appropriate units and within the context of a specific population.

Essential question: When two probability models have the same mean, how can standard deviation change the decision?

DOK-3 skill: 4.B · **FRQ pattern:** `same-mean-different-sd-which-parameter-answers-the-question`

DOK rationale: Part (c) weighs competing claims using the day's numerical or graphical evidence and explains a limitation or consequence; the judgment requires linked reasoning beyond a calculation.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which number or visible feature supports your claim?
- Why does the other claim go beyond that evidence?
- What would you tell someone who chose the other claim?

[WARN] LOOK FOR

- A choice with copied numbers but no explanation of how they support it.
- Treating a model or average as a guaranteed individual outcome.

SCORING PART (c) — E / P / I

Expected elements

- **reasoning-1** (required) — Chooses A using standard deviations 3 versus 12 minutes
- **reasoning-2** (required) — Connects 0/8 versus 2/8 above 40 to the customer's deadline
- **reasoning-3** (required) — Explains that equal means do not imply equal variability or late risk

E	Includes all three required reasoning elements above, with correct contextual evidence.
P	Includes at least one substantive correct reasoning link above, but omits or misstates another required element.
I	Includes no substantive correct reasoning link above; an unsupported choice or copied number alone is insufficient.

Common mistakes

- Treating standard deviation as minutes added to every delivery
- Claiming every Route B time is exactly 12 minutes from its mean
- Using equal means alone to infer equal chances of missing the deadline

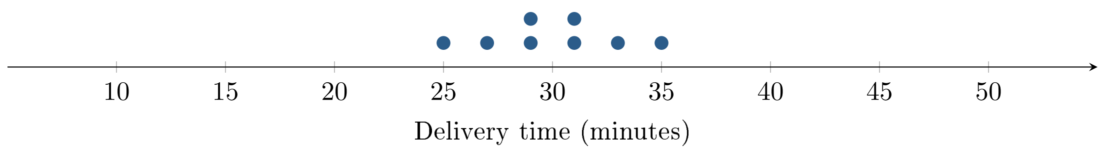
[STAR] THE PROBLEM — annotated

Suppose a dispatcher models delivery time using eight equally likely outcomes for each route, shown in the dotplots. An analyst reports that both routes have mean 30 minutes, with $\sigma_A = 3$ minutes and $\sigma_B = 12$ minutes.

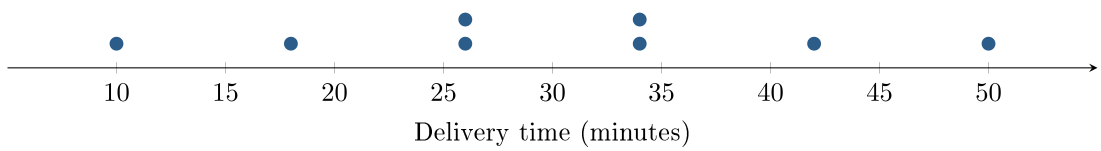
The dispatcher says, *The routes have the same mean, so it does not matter which one a customer gets.* A customer has 40 minutes before an appointment and places a high cost on being late.

For the eight outcomes, the sums of squared distances from 30 are 72 for Route A and 1,152 for Route B.

Route A



Route B



FIRST TAKE — one sentence before you work the parts

Which route would you assign to this customer? Cite one number or dotplot feature.
Key: Not graded. Route B's very fast outcomes may attract some students before deadline risk is made explicit.

[BOOK] MEAN AND STANDARD DEVIATION (keep on the page)

For a discrete random variable, $\mu_X = \sum xP(X = x)$ and $\sigma_X = \sqrt{\sum(x - \mu_X)^2 P(X = x)}$. For n equally likely outcomes, divide the sum of squared distances from the mean by n, then take the square root. The mean is a long-run average; the standard deviation describes a typical distance from it, in the variable's units. Add probabilities of outcomes that miss a deadline.

DOK 1 **(a)** Count the outcomes longer than 40 minutes for each route.

Key: Route A has no outcomes longer than 40 minutes. Route B has two: 42 and 50 minutes.

DOK 2 **(b)** Use the given sums of squared distances to verify both population standard deviations. Interpret what one of them means in minutes.

Key: $\sigma_A = \sqrt{72/8} = 3$ minutes; $\sigma_B = \sqrt{1152/8} = 12$ minutes. Route A's modeled delivery times typically differ from their 30-minute mean by about 3 minutes.

DOK 3 **(c)** Which route fits this customer's priority? Use both standard deviations and the chance of exceeding 40 minutes. Explain why the dispatcher's mean-only comparison can mislead this customer. Write 3–4 sentences.

Key: Choose Route A: its standard deviation is 3 minutes rather than 12, so its modeled times vary less around the same 30-minute mean. No A outcome exceeds 40 minutes, compared with $2/8 = 25\%$ for B. Equal means do not imply equal chances of being late, so the dispatcher's comparison hides the difference this customer cares about. This conclusion uses the eight equally likely outcomes in the stated model.